Fitbit Data Analysis Task

Overview

The aim of this task is to test and refine your teamworking skills before the assessed group projects commence in January 2027.

To this end we are setting you a series of tasks to perform in your 'Training Assessment Groups (TAGs)'. These were assigned in Week 2 at the Block Release and relate to your tutor groups assigned last year. In your TAG you are to get a full practice run of what will be expected in the 'Assessed Group Project (AGPs)'. In this document are all the details you will need to work through this assignment, including your team working expectations, personal reflections and scientific content.

The problem is this: you have been given a folder of data from a wearable device, specifically a Fitbit dataset which requires an analysis dashboard to be completed and tested. You are to work together to implement the dashboard. The dataset is provided on the DATA2007 Moodle page. You shall create a repository to log your contributions using Markdown files produced weekly. Your team is to complete the dashboard as a deliverable and to produce a retrospective of how you worked as a team and your findings from using this dashboard in the form of a video presentation.

The Fitbit Data

"This dataset was generated by respondents to a distributed survey via Amazon Mechanical Turk between 03.12.2016-05.12.2016. Thirty eligible Fitbit users consented to the submission of personal tracker data, including minute-level output for physical activity, heart rate, and sleep monitoring. Individual reports can be parsed by export session ID (column A) or timestamp (column B). Variation between output represents the use of different types of Fitbit trackers and individual tracking behaviours/preferences." - Description from the data source.

The dataset contains a collection of CSV files, quite nicely formatted with very few missing or redundant data streams. As a first pass through, examine the data given and perform some basic statistical and numerical analysis, and some data cleaning. Each file represents a different measurement type with different attributes, some are aggregated over hours, some minutes and seconds. Top tip: use something nice like R or Pandas to collectively explore the dataset yourselves.

As a TAG you must decide what you want to do to this data and how you would present this in a useful dashboard. Assume that the users are all athletes in a friend group and want to see how all members in the group are exercising.

You can download the data here.

Creating your Dashboard

Think about your end-user requirements when creating a dashboard to allow a user to explore the data. These shall be formally documented by your group in a single markdown file saved to a shared. In week 4, training material will be provided to help you think through and format data requirements. You should aim to have between 25 and 40 requirements in total. Top tip: use the Easy Approach to Requirements Syntax to structure your requirements.

Once you have settled on your requirements, proceed to construct your dashboard. You may use any programming technique you wish to do this. You can set up a database for the CSV files or simply write a web interface to the CSV files, your TAG must make this decision and record your justification for this and for the architecture of your system. Upon implementation remember to

ensure your implementation aligns with your requirements. Top tip: remember to unit test your code as you go along to avoid cascading errors.

Documentation and Retrospective

As a group you will be expected to produce:

1. 500 words initial data exploration summary
2. A minimum of 25 user requirement
3. Dashboard implementation and code in github repository
4. 6 minute retrospective video of your dashboard and project summary findings

As an individual you are to be expected to produce:

1. Repository of weekly personal reflections logging your contributions for the week and your plan for the coming week
2. Written up as a markdown file each week
3. A peer review of each of the group video retrospective videos summarised as class findings and how you will improve for the AGP.